

Homework — 1.1.3 Input, Output and Storage Devices

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.1.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. A supermarket self-checkout must (a) read the barcode on each product, (b) show the running total to the customer, and (c) produce a paper receipt. State a suitable device for **each** of the three tasks. **[3]** (AO2)

- (a): _____
- (b): _____
- (c): _____

2. Describe how data is stored and read on an **optical** disc such as a DVD. **[2]** (AO1)

3. A laptop manufacturer replaces a magnetic **HDD** with a **flash (SSD)** drive. State **two** advantages and **one** disadvantage of this change. **[3]** (AO1)

Advantages: _____

Disadvantage: _____

4. A museum needs to **archive** several terabytes of footage that will rarely be accessed again, at the **lowest possible cost**. Recommend a suitable storage medium and **justify** your choice against the requirements. [3] (AO2)

5. Explain the difference between **RAM** and **ROM** in terms of volatility and what each is used for. [2] (AO1)

6. A student saves their coursework to an online (cloud) service instead of the school's local file server. State what is meant by **virtual storage**. [1] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

7. State **two** differences between **RISC** and **CISC** processors. [2] (*retrieval from 1.1.2*)

8. Explain why a **GPU** is well suited to applying the same calculation to a large data set. [2] (*retrieval from 1.1.2*)

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9. State the **full name** and **role** of the **MDR**. [2] (*retrieval from 1.1.1*)

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